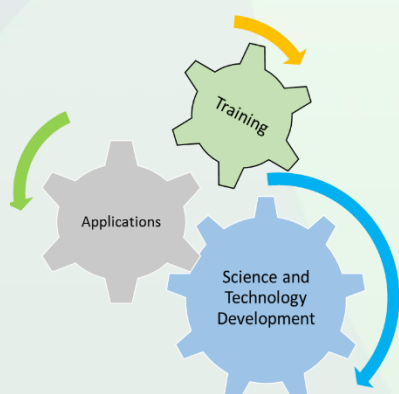




A warm welcome to Centre of Excellence for Biopharmaceutical Technology!

The **Center of Excellence for Biopharmaceutical Technology (CBT)** was established at IIT Delhi in 2015 by the Department of Biotechnology, Government of India, in recognition of the importance of Biotechnology for India, particularly that of producing **affordable biotech therapeutics**.

The vision of CBT is to deliver innovation in biopharmaceutical technology to effectively address the challenges faced by the Indian biotech industry and thereby assist in the “**Make in India**” initiative by making India the **global hub** of manufacturing economical, safe and efficacious therapeutics.



CBT aspires to achieve this tall objective by providing a foundation of scientific and technology development to create **novel technologies**, engage with the biotech industry to **translate** these into applications, and finally to offer short term training courses to industry, academia, and regulatory agencies to facilitate **creation of an ecosystem** that delivers **affordable biotech therapeutics** to India and to the world.

CBT is bringing forth a **world class training program** that brings together leaders from across the world to come together and share the best practices and cutting edge technologies on a diverse set of topics:

Elucidating the Guidelines on Similar Biologics for India

Proteomics for Characterization of Biotherapeutics

Design of Experiments

Process Optimization and Scale-Up with Single-Use Bioreactors

Continuous Processing for production of Biopharmaceuticals

Multivariate Data Analysis for Bioprocessing Data

Formulation and Stability of Biotherapeutic Product



Prof. Anurag S. Rathore
Coordinator, CBT, IIT Delhi



Prof. James Gomes
Co-Coordinator, CBT, IIT Delhi



Centre of Excellence
Biopharmaceutical
Technology



THANK YOU SPONSORS!



सत्यमेव जयते

Department of Biotechnology



MERCK





CBT COURSE SERIES: INAUGURAL SESSION

Date: 12th December 2018

Time: 08.30-10.00

Venue: LH 418

AGENDA

- | | |
|-------------|--|
| 08.30-09.00 | Registration and Breakfast |
| 09.00-09.15 | Welcome Address
<i>(Prof. V. Ramgopal Rao, Director, IITD)</i> |
| 09.15-09.30 | Introduction to the course
<i>(Prof. Anurag S. Rathore, IITD)</i> |
| 09.30-09.45 | Academia-Industry-Government Partnerships
<i>(Dr. S. Eswara Reddy, DCGI)</i> |
| 09.45-10.00 | Need of Training in Biopharmaceutical Technology
<i>(Dr. M. K. Bhan, Ex-Secretary, DBT)</i> |
| 10.00-10.30 | Tea Break |



Prof. V. Ramgopal Rao



Prof. Anurag S. Rathore



Dr. S. Eswara Reddy



Dr. M. K. Bhan



ELUCIDATING THE GUIDELINES ON SIMILAR BIOLOGICS FOR INDIA (12th to 14th December, 2018; Venue: LH 418)

DESCRIPTION: Regulatory approval is the desired endpoint of all development and commercialization activities in the biopharmaceutical industry. A critical aspect of similar biologics development is having comprehensive regulatory guidelines. Indian regulatory authorities published the Guidelines on Similar Biologics in 2012 with a revision in 2016. This proposed three-day course aims to strengthen our regulators and the industry on the various aspects of the Guidelines: process and analytical development, preclinical and clinical development, and regulatory and Good Manufacturing Practices (GMP) issues that relate to development and manufacturing of biopharmaceutical products. The audience is expected to consist primarily of Indian regulators and other stakeholders that play a key role in the approval process. Limited participants from the Indian biotech industry will also be accommodated.

Outline

Day 1: Process and Analytical Development (12th December 2018)

- 08.30-09.00 Registration and Breakfast
- 09.00-10.00 Inaugural Session
- 10.00-10.30 Tea Break
- 10.30-11.30 Key aspects of upstream process development
(*Prof. James Gomes, IITD*)
- 11.30-12.30 Key aspects of downstream process development
(*Prof. Shuichi Yamamoto, Yamaguchi University, Japan*)
- 12.30-13.30 Scale-up of bioprocessing unit operations
(*Prof. Anurag S. Rathore, IITD*)
- 13.30-14.30 Lunch Break
- 14.30-15.30 Key aspects of analytical development
(*Prof. Andras Guttman, University of Debrecen, Hungary*)
- 15.30-16.00 Tea Break
- 16.00-16.30 **Case Study:** Protein A and related molecules for antibody purification (*Dr. Richard C. Willson, University of Houston, USA*)
- 16.30-17.00 **Case Study:** Key considerations in biologics manufacturing
(*Prof. E.K. Lee, Hanyang University, South Korea*)
- 17.00-17.15 Discussion and Wrap Up



Prof. Anurag S. Rathore



Prof. James Gomes



Prof. Shuichi Yamamoto



Prof. Andras Guttman



Dr. Richard Willson



Prof. E. K. Lee



**Day 2: GMP Manufacturing and Regulatory Approval Process
(13th December 2018)**

- 08.30-09.00 Breakfast
- 09.00-09.45 An overview of the regulatory approval process
(*Dr. Chandrashekar Ranga, CDSCO*)
- 09.45-10.30 Key deficiencies in regulatory filings for biosimilars
(*Dr. V. Siva Reddy, BSU, RCGM*)
- 10.30-11.00 Tea Break
- 11.00-12.30 Key issues in GMP manufacturing of biotherapeutic products
(*Dr. Varnika Roy, GlaxoSmithKline, USA*)
- 12.30-13.30 Lunch Break
- 13.30-14.30 Process monitoring and validation for biopharma
manufacturing (*Prof. Anurag S. Rathore, IITD*)
- 14.30-15.15 **Case Study:** GMP manufacturing and globalization strategy
for recombinant Hepatitis-B vaccine
(*Prof. E.K. Lee, Hanyang University, South Korea*)
- 15.15-15.45 Tea Break
- 15.45-16.30 Comparability approaches for cell line and process
changes - a perspective (*Dr. M. S. Ramakrishnan, Biocon*)
- 16.30-17.00 Discussion and Wrap Up



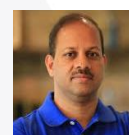
Dr. Chandrashekar Ranga



Dr. V. Siva Reddy



Dr. Varnika Roy



Prof. Anurag S Rathore



Prof. E.K. Lee

**Day 3: Preclinical and Clinical Evaluation and Pharmacovigilance
(14th December 2018)**

- 08.30-09.00 Breakfast
- 09.00-10.30 Critical quality attributes of a biotherapeutics
(*Dr. Varnika Roy, GlaxoSmithKline, USA*)
- 10.30-11.00 Tea Break
- 11.00-12.30 Preclinical evaluation for biosimilars
(*Dr. Sesikaran, RCGM*)
- 12.30-13.15 Lunch Break
- 13.15-14.15 Clinical evaluation for biosimilars
(*Dr. Nilima Kshirsagar, ICMR, New Delhi*)
- 14.15-15.45 Data requirements for MAA
(*Dr. M.S. Ramakrishnan, Biocon*)
- 15.45-16.15 Tea Break
- 16.15-17.15 Pharmacovigilance and ADR reporting
(*Dr. Sunita Singh, NDMC Hindu Rao Hospital*)
- 17.15-17.30 Discussion and Wrap Up



Dr. M.S. Ramakrishnan



Dr. Sesikaran



Dr. Nilima Kshirsagar



Dr. Sunita Singh



FORMULATION AND STABILITY OF BIOTHERAPEUTIC PRODUCTS (12th December 2018; Venue: LH 421)

DESCRIPTION: Stability of biotherapeutics is paramount to their successful commercialization. Proteins are naturally prone to degradation, with pathways of aggregation and fragmentation fairly common. Creating optimal formulations for these products is non-trivial. Not only we need fundamental understanding of the various degradation pathways and their kinetics, one also has to be aware of the intellectual property on the topic. This course will aim to share both the fundamentals of the subject as well as the best practices from the industry.

OUTLINE:

- | | | |
|-------------|---|---|
| 08.30-09.00 | Registration and Breakfast | |
| 09.00-10.00 | Inaugural Session | |
| 10.00-10.30 | Tea Break | |
| 10.30-11.30 | Chemical and physical degradation of proteins
(<i>Dr. Satish Singh, Lonza</i>) | 
Dr. Satish Singh |
| 11.30-12.15 | Protein analysis and stability of protein
(<i>Dr. Karthik Ramani, Biocon</i>) | |
| 12.15-13.15 | Rational development of protein formulations
(<i>Dr. Satish Singh, Lonza</i>) | 
Dr. Karthik Ramani |
| 13.15-14.00 | Lunch | |
| 14.00-15.00 | Protein product quality related factors and immunogenicity
(<i>Dr. Karthik Ramani, Biocon</i>) | 
Dr. Avinash Mishra |
| 15.00-16.00 | Protein drug product manufacturing process, including lyophilization (<i>Dr. Satish Singh, Lonza</i>) | |
| 16.00-16.15 | Break | |
| 16.15-16.45 | Kinetics of aggregation and fragmentation of monoclonal antibody products (<i>Mr. Rohit Bansal, IITD</i>) | |
| 16.45-17.15 | Case Study: Simulation based design of novel stabilizers for protein drugs (<i>Dr. Avinash Mishra, Novo Informatics</i>) | |
| 17.15-18.00 | Special topics in biotherapeutics development (<i>Dr. Satish Singh, Lonza</i>) | |
| 18.00-18.15 | Discussion and Wrap-up | |



**CONTINUOUS PROCESSING FOR PRODUCTION OF BIOPHARMACEUTICALS:
CONCEPTS, DEMO AND CASE STUDIES
(12th December, 2018; Venue: LH 416)**

DESCRIPTION: The merits of continuous processing over batch processing are well known in the manufacturing industry. Continuous operation results in shorter process times due to omission of hold steps, higher productivity due to reduced shut down costs and lowers labor requirement. This course will address the topic of continuous processing with a focus on production of biotech therapeutics. Basic concepts that provide a foundation for this exercise will be presented together with a few case studies that demonstrate process integration. A demo of continuous process will also be displayed.

OUTLINE:

- 08.30-09.00 Registration and Breakfast
- 09.00-10.00 Inaugural session
- 10.00-10.30 Tea Break
- 10.30-11.00 Continuous Processing (*Prof. Anurag S. Rathore, IITD*)
- 11.00-12.00 Introduction to Continuous Processing Unit Operations
(*Dr. Rick Morris, Pall Corporation*)
- 12.00-13.00 Enabling Technologies: Multicolumn chromatography
(*Dr. Rick Morris, Pall Corporation*)
- 13.00-13.45 Lunch
- 13.45-15.15 Demo – Continuous Processing
- 15.15-15.45 **Case Study 1:** Use of continuous flow inverted reactor for continuous protein refolding and precipitation
(*Mr. Nikhil Kateja, IITD*)
- 15.45-16.15 Break
- 16.15-16.45 **Case Study 2:** Non-protein A purification platform for continuous processing of monoclonal antibody therapeutics
(*Ms. Anamika Tiwari, IITD*)
- 16.45-17.15 **Case Study 3:** Continuous downstream process for production of GCSF
(*Mr. Nikhil Kateja, IITD*)
- 17.15-17.45 **Case Study 4:** BioSMB (*Dr. Masilamani Selladurai, Pall Corporation*)
- 17.45-18.00 Discussion and Wrap-up



Prof. Anurag S. Rathore



Dr. Rick Morris



Dr. M. Selladurai



DESIGN OF EXPERIMENTS (13th December, 2018; Venue: LH 416)

DESCRIPTION:

Quality by Design (QbD) and Design Space estimation, initiated by ICH, are moving to the phase of establishing standardized procedures and definitions within biopharmaceutical companies. The challenge is to extract a mathematically useful description that matches the flexibility and risk control described in the ICH Q8 guidelines. Design of Experiments (DOE) is required for a good and reliable Design Space description. However, most of the set-points derived from DOE are overoptimistic and do not consider risk of failure. A design space can be an irregular multidimensional region or a strict Proven Acceptable Range (PAR) for Critical Process Parameters (CPP's). This course will present the fundamentals of DOE and illustrate how to derive a multidimensional Design Space. Hands-on exercise is an integrated part of the course which let you experience how DOE and Design space estimation are carried out.

OUTLINE:

- 08.30-09.00 Breakfast
- 09.00-10.15 Introduction to DOE
(*Dr. David Wang, Vaibhav Patil, Sartorius Stedim*)
- 10.15-10.45 Break
- 10.45-11.15 Role of DOE in Biopharmaceutical Development
(*Prof. Anurag S. Rathore, IITD*)
- 11.15-12.45 DOE for screening and hands-on exercises
(*Dr. David Wang, Vaibhav Patil, Sartorius Stedim*)
- Increasing soluble and insoluble protein from *E. coli* production of API
- 12.45-13.15 **Case Study:** DOE based screening of media components
(*Mr. Neelesh Gangwar, IITD*)
- 13.15-14.00 Lunch
- 14.00-15.30 DOE for optimization and Hands-on exercises
(*Dr. David Wang, Vaibhav Patil, Sartorius Stedim*)
- Optimization of a fermentation process
 - Media Optimization
- 15.30-16.00 Break
- 16:00-17.00 **Case Study:** Analysis and validation of statistical models during process characterization (*Mr. Solomon Alva, Biocon*)
- 17.00-18.00 Find robust set-point for design space using DOE and Monte Carlo simulations; DOE on stability testing
(*Dr. David Wang, Mr. Vaibhav Patil, Sartorius Stedim*)
- 18.00-18.15 Discussion and Wrap-up



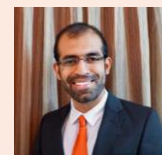
Dr. David Wang



Mr. Vaibhav Patil



Prof. Anurag S. Rathore



Mr. Solomon Alva



PROTEOMIC CHARACTERIZATION OF BIOTHERAPEUTICS: CONCEPTS AND CASE STUDIES

(13th December 2018; Venue: LH 421)

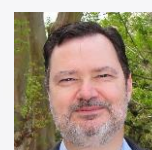
DESCRIPTION: Analytical characterization is the backbone of establishing comparability, arguably the most critical step in development and commercialization of biotherapeutic products. Major advancements have occurred in the past few years with respect to both development of novel, high resolution analytical tools as well as of novel approaches. Of these, arguably the most significant is the emergence of mass spectrometry as the tool of choice for identification and quantification of a biotherapeutic. In this course, we will showcase the novel applications of this tool with the aim to demonstrate the power and dexterity of this tool.

OUTLINE:

- 08.30-09.00 Breakfast
- 09.00-09.30 Introduction to biopharmaceutical analysis
(*Prof. Anurag S. Rathore, IITD*)
- 09.30-10.30 Biophysical methods of antibody characterization
(*Dr. Richard C. Willson, University of Houston, USA*)
- 10.30-11.00 Break
- 11.00-12.00 Data analysis for proteomic applications
(*Mr. Saurabh Nagpal, Agilent Technologies*)
- 12.00-12.45 Challenges in intact protein & peptide mapping analysis
(*Dr. Ashish Pargaonkar, Agilent*)
- 12.45-13.45 Lunch
- 13.45-14.30 Cutting edge applications for automated glycan sequencing and multilevel analysis of biologics (including CE, LC, MS)
(*Prof. Andras Guttman, University of Debrecen, Hungary*)
- 14.30-15.00 Challenges in analysis of post translational modifications and host cell proteins (*Dr. Ashish Pargaonkar, Agilent*)
- 15.00-15.30 **Case Study 1:** Analytical comparability of Rituximab and Herceptin biosimilars (*Dr. Srishti Joshi, IITD*)
- 15.30-16.00 Break
- 16.00-17.00 Biosimilarity assessment focusing on glycosimilarity index
(*Prof. Andras Guttman, University of Debrecen, Hungary*)
- 17.00-17.30 **Case Study 2:** LC-MS based PMF analysis for biosimilarity assessment (*Ms. Neh Nupur, IITD*)
- 17.30-17.45 Discussion and Wrap-up



Prof. Anurag S. Rathore



Dr. Richard Willson



Mr. Saurabh Nagpal



Prof. Andras Guttman



Dr. Ashish Pargaonkar



MULTIVARIATE DATA ANALYSIS FOR BIOPROCESSING DATA: CONCEPTS AND CASE STUDIES (14th December, 2018: Venue: LH 416)

DESCRIPTION: Biopharma and biotech manufacturing today involves larger and larger masses of data. Correct management of these data can give us valuable insights to process development as well as production, help us understand the progress of a batch and also point us in the right direction to troubleshoot. By applying state-of-the-art data analysis technologies in the biopharma production the time for fault detection and diagnosis in production can be significantly reduced. In one recent example, a company identified the cause of a cell culture problem about a month earlier than it otherwise might have. For that biologic product, making the fix early and not losing that month saved \$2.4 million. The course will present fundamental theory and practice of Multivariate Data Analysis (MVDA) within the framework of PAT to improve quality and throughput, as well as hands-on exercises on how MVDA should be used in process development and production.

OUTLINE:

08.30-09.00 Breakfast

09.00-10.30 Introduction to MVDA (*Dr. David Wang, Sartorius Stedim*)

Principal Component Analysis (PCA)

- Explanation, Score plot & Loading plot
- Diagnosis of observations, variables and model

10.30-11.00 Break

11.00-11.45 Role of MVDA in Biopharmaceutical Development

(*Prof. Anurag S. Rathore, IITD*)

11.45-12.45 Hands on exercises (*Dr. David Wang, Vaibhav Patil, Sartorius Stedim*)

- Main Steps of data analysis with SIMCA
- **Case study:** Multi-scale cell culture data overview
- **Case study:** Patients healthcare data

12.45-13.30 Lunch

13.30-14.00 **Case Study:** Multivariate analysis of mass spectrometry data for biosimilars (*Mr. Shantanu Banerjee, IITD*)

14.00-15.15 Partial least squares (PLS) and orthogonal partial least squares (OPLS) (*Dr. David Wang, Vaibhav Patil, Sartorius Stedim*)

- **Case study:** Developing Calibration Model in Spectroscopy

15.15-15.45 Break

15.45-17.15 How to build batch models (*Dr. David Wang, Mr. Vaibhav Patil, Sartorius Stedim*)

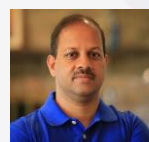
- Batch evolution model & batch level model
- **Case study:** Modelling a fermentation process for development, monitoring, and control

17.15-17.45 PAT implementation for protein refolding: GCSF as a case study (*Mr. Vishwanath Hebhi, IITD*)

17.45-18.00 Discussion and Wrap-up



Dr. David Wang



Prof. Anurag S. Rathore



Mr. Vaibhav Patil



PROCESS OPTIMIZATION AND PROCESS SCALE-UP WITH SINGLE-USE BIOREACTORS

(14th December 2018: Venue: LH 421)

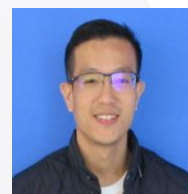
DESCRIPTION: Single-Use Bioreactors are becoming increasingly popular and replacing traditional stainless steel bioreactors as the main choice for many manufacturers. However, many Single-Use Bioreactors adopt flexible designs that may diverge from classical geometries. The adoption of Single-Use technologies and disposables has its advantages but also introduces new concerns in relation to their potential impact on process variables. The by-products from the breakdown of these disposables during high temperature/solvent processing may have an impact on cell growth, productivity, quality and safety. This course will aim to share various principles that facilitate process transfer and scale-up from a traditional bioreactor to a Single-Use Bioreactor.

OUTLINE:

- 08.30-09.00 Breakfast
- 09.00-09.45 Introduction to single-use technologies for biopharmaceutical manufacturing industries
(*Prof. E.K. Lee, Hanyang University, Korea*)
- 09.45-10.45 Process Optimization and Process Scale-Up with Single-Use Bioreactors (*Dr. Julius Ang, Merck*)
- 10.45-11.15 Break
- 11.15-12.45 Control of bioreactors (*Prof. James Gomes, IITD*)
- 12.45-13.15 Next Generation Bioprocessing – Opportunities in Upstream Process Intensification (*Dr. Julius Ang, Merck*)
- 13.15-14.00 Lunch
- 14.00-15.00 Optimize your sampling plan with Single-use technology
(*Mr. Ashok Kumar N., Merck Life Sciences, India*)
- 15.00-15.30 **Case Study 1:** Advanced process control schemes for bioreactors
(*Prof. James Gomes, IITD*)
- 15.30-16.00 Break
- 16.00-16.30 **Case Study 2:** Effect of media components on microbial fermentation (*Mr. Jashwant Kumar, IITD*)
- 16.30-17.30 **Case Study 3:** Effect of media components on mammalian cell culture (*Mr. Neelesh Gangwar, IITD*)
- 17.30-17.45 Discussion and Wrap-up



Prof. E. K. Lee



Dr. Julius Ang



Prof. James Gomes



Mr. Ashok Kumar N.



REGISTRATION FEES DETAILS

No. of Days	Indian Academic	Indian Industry Participant	Foreign Academic	Foreign Industry Participant
One Day Course	5000 INR+ 18% GST	7000 INR+ 18% GST	100 USD +18% GST	200 USD + 18% GST
Three Day Course	15000 INR+ 18% GST	20000 INR + 18% GST	300 USD +18% GST	500 USD+18% GST

Please note the fees mentioned above for one day course is not applicable for MVDA & DOE courses.

For DOE and MVDA Courses:

No. of days	Indian Academic	Indian Industry Participant	Foreign Academic	Foreign Industry participant
One Day	8000 INR+ 18% GST	10000 INR + 18% GST	100 USD +18% GST	200 USD +18% GST

LODGING AND BOARDING: Accommodation can be booked in nearby hotels/ guesthouses directly. For information, **please refer Annexure 'A'** (Last page of the Brochure).

MODE OF PAYMENT FOR REGISTRATION: The registration fee should be paid only by NEFT transfer to the following account details mentioned below. The scanned registration form along with registration fee receipt should be sent by e-mail to Course Coordinator **Prof. Anurag S. Rathore (IIT Delhi)**, at coe.biopharma.course@gmail.com

Beneficiary Name: Foundation for Innovation and Technology

Bank Name: State Bank of India

Account #:10773571968

Branch Name: SBI, IIT Delhi Hauz Khas, New Delhi-110016

IFS Code: SBIN0001077

MICR: 110002156

Swift No. SBI NIN BB 547

Bank Code: 1077



REGISTRATION FORM

Name: Prof./Dr./Mr./Ms.:

Designation:

Department:

University/Company:

Address:

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Email ID:

Mobile/Phone:

Name of the Course/Courses opted for:

.....

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Transaction ID (NEFT):

Place:

Date:

Signature of the Applicant:

For any queries, please contact coe.biopharma.course@gmail.com



ANNEXURE-A

Hotel Name	Distance from IITD	Room Type	Rate (approx.) (Ex GST)	Address	Contact Details
The Ashton Sarovar Portico	3.5 Km	Single Double	Rs. 4700 Rs. 5400	C-2, Green Park Extension, New Delhi-110016	Website: www.astonhotels.com Landline: 011 4683 3333 Email: gppo@sarovarhotels.com
Mint	4.7 Km	Single Double	Rs. 3500 Rs. 4000	A1/11 Safdarjung Enclave N D, 110029, Delhi	Website: www.staymint.com/hotels/new-delhi/hotel-mint-safdarjung/ Mobile: 9818092555 Email: booking@staymint.com
The Visaya	4.9 Km	Single Double	Rs. 5000 Rs. 5500	N 82 Panchshila Park, 110017, Delhi	Website: www.thevisaya.com Mobile: 9560431112 Email: info@thevisaya.com
Rainbow Residency	5.3 Km	Single Double	Rs. 3000 Rs. 4500	G-52, Green Park, 110016, Delhi	Website: www.rainbowresidency.in Mobile: 7838666388 Email: mail@rainbowresidency.in